PALADINMEDIA — WEEK 7 REPORT

Sprint-1 Feature Testing & Review (Test Log, Retrospective & Backlog Update)

Project Title:	PaladinMedia — Post-Quantum Media Steganography System
Milestone Window:	WEEK 7 | Sprint-1 Testing & Review | 7 – 9 October 2026
Individual Contributor / Owner:	Rohit Gautam (Individual Project)
Current Milestone Status:	**Completed & 100% Verified (99/99 Pytest Suite Passing)**

1. Executive Summary & Sprint-1 Testing Context

During Week 7 (7 – 9 October 2026), the scheduled testing and review cycle for Sprint-1 was executed. As an individual engineering project, the review and retrospective activities were conducted through a rigorous self-audit and automated test validation rather than team ceremonies. All core Sprint-1 deliverables—comprising the Secret Payload Serializer, AES-256-GCM Cryptographic Vault, Reed-Solomon Error Correction Encoder, and the initial Carrier Wavelet Integration—were subjected to exhaustive automated unit, integration, and security verification. A total of 99 automated test cases were executed with a 100% pass rate (0 failures, 0 regressions), successfully verifying complete fail-closed security, mathematical invertibility, and byte-exact media recovery.

2. Sprint-1 Feature Testing Log

The following test log details the formal verification scenarios executed against the Sprint-1 feature baseline, mapping test cases to their respective components, execution types, results, and empirical validation metrics:

Test ID	Feature / Module	Test Scenario Description	Type	Result	Verified Metric / Behavior
TL-01	SecretSerializer	Binary framing roundtrip with JSON metadata & CRC-32	Unit	**PASS**	Exact byte match
TL-02	SecretSerializer	CRC-32 checksum mismatch & corrupted header rejection	Security	**PASS**	SerializationError
TL-03	AESGCMProvider	NIST known answer vector encryption & decryption	Unit	**PASS**	Exact plaintext
TL-04	AESGCMProvider	Single-bit ciphertext tampering	Security	**PASS**	Fail-closed (0 leak)

		& tag modification			
TL-05	AESGCMProvider	Mismatched Associated Authenticated Data (AAD)	Security	**PASS**	AuthenticationError
TL-06	ReedSolomonEncoder	GF(2^8) RS(255, 223) encoding with 32 parity bytes	Unit	**PASS**	32B parity generated
TL-07	ReedSolomonEncoder	Correct symbol errors within t=16 threshold	Integration	**PASS**	100% error corrected
TL-08	ReedSolomonEncoder	Catastrophic error injection (>16 symbols corrupted)	Security	**PASS**	ECCDecodeError
TL-09	CapacityCalculator	Hard-gate capacity verification at 8,176 B ceiling	Unit	**PASS**	PayloadCapacityExceeded
TL-10	HaarDWT2D	One-level 2D Haar DWT / IDWT floating-point invertibility	Unit	**PASS**	Max error = 0.000000
TL-11	CarrierHeader	16-byte fixed canonical framing codec (b'PLDH')	Unit	**PASS**	Pack/unpack identity
TL-12	MediaStegoPipeline	Full roundtrip: Secret -> Stego PNG -> Reload -> Recover	E2E	**PASS**	PSNR: 75.89 dB, BER: 0.0

3. Individual Sprint Retrospective (Self-Review)

As an individual project, the sprint review and retrospective is structured as a dedicated engineering self-assessment focusing on technical outcomes, obstacles overcome, and process refinements:

Retrospective Category	Detailed Engineering Observations	Action Items / Future Impact
What Went Well (Successes & Strengths)	• Strict unidirectional pipeline (Serialize -> Encrypt -> ECC -> Embed) established without architectural leakage. • Absolute fail-closed cryptographic implementation verified: zero plaintext released under all tampering tests. • Zero bit errors (BER = 0.0) achieved across 8-bit RGB PNG disk reload using integer-parity modulation on Haar coefficients. • High carrier fidelity achieved: 75.89 dB PSNR and	Maintain frozen cryptographic contracts and preserve existing unit test suite across future phases.

	1.000000 SSIM on baseline carrier images.	
Challenges Faced (Obstacles & Resolutions)	• Theoretical vs Usable Capacity Gap: Initial assumption of 8,192 B capacity collided with real protocol overheads. Resolved by modeling explicit capacity accounting (8,176 B protected, ~7,088 B usable secret). • Oversized Payload Handling: Large images (e.g. 791 KB) triggered capacity failures. Resolved by strictly enforcing the capacity hard-gate (PayloadCapacityExceededError) before transform execution.	Incorporate adaptive lossless compression (zlib) and multi-carrier transport protocol (MCPP) for large payloads.
Key Learnings & Process Refinements	• Automated test-driven verification eliminated regressions early. • Separating classical signal processing from future neural research prevented premature scope creep. • Disk I/O (PNG encode/decode) accounts for >75% of total roundtrip latency (~102 ms out of 134 ms total).	Optimize image buffer operations in memory; prepare data tensors cleanly for Phase 3 neural engine integration.

4. Updated Project Backlog

Following the Sprint-1 review and retrospective, the master project backlog has been formally updated. Sprint-1 work items are finalized as DONE, while upcoming Sprint-2 and Phase 3 items are groomed and prioritized:

Ticket ID	Task / User Story	Sprint / Phase	Priority	Status	Resolution / Milestone Note
PDM-S1-001	Freeze module contracts and wire specifications	Sprint-1	**P0**	**Done**	Contracts frozen
PDM-S1-002	Implement AESGCMProvider (AES-256-GCM)	Sprint-1	**P0**	**Done**	NIST vectors pass
PDM-S1-003	Fail-closed tamper handling and zero-leak verification	Sprint-1	**P0**	**Done**	All negative tests pass
PDM-S1-004	Implement ReedSolomonEncoder (RS GF(2^8))	Sprint-1	**P0**	**Done**	Corrects t=16 symbols
PDM-S1-005	Implement SecretSerializer with CRC-32 integrity	Sprint-1	**P0**	**Done**	b'PLDS' framing

PDM-S1-006	Sprint-1 Cryptographic integration test suite	Sprint-1	**P0**	**Done**	99/99 pytest green
PDM-S2-001	Haar DWT/IDWT 2D transform engine & capacity accounting	Sprint-2	**P0**	**Done**	Max error = 0.0
PDM-S2-002	16-byte fixed CarrierHeader codec (b'PLDH')	Sprint-2	**P0**	**Done**	Anchor positions 0..127
PDM-S2-003	Non-circular DeterministicMaskGenerator via SHA-256 PRNG	Sprint-2	**P0**	**Done**	Seed-reproducible
PDM-S2-004	Baseline Integer-Parity Embedding & Extraction engine	Sprint-2	**P0**	**Done**	BER = 0.0, PSNR = 75.89 dB
PDM-S2-005	Phase 2 Experimental Program Execution (E03 to E11)	Sprint-2	**P0**	**Done**	Capacity frontier mapped
PDM-S3-001	ACS-Sub-INN Neural Invertible Flow Architecture	Sprint-3 (Phase 3)	**P1**	**Planned**	Pending Phase 3 kickoff
PDM-S3-002	Context Predictor CNN & Adaptive Gain Network	Sprint-3 (Phase 3)	**P1**	**Planned**	Wavelet feature input
PDM-S3-003	Straight-Through Estimator (STE) Quantization Noise Layer	Sprint-3 (Phase 3)	**P2**	**Planned**	Lossy container training